Diagnosis and management of uncomplicated urinary tract infections.

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Most uncomplicated urinary tract infections occur in women who are sexually active, with far fewer cases occurring in older women, those who are pregnant, and in men. Although the incidence of urinary tract infection has not changed substantially over the last 10 years, the diagnostic criteria, bacterial resistance patterns, and recommended treatment have changed. Escherichia coli is the leading cause of urinary tract infections, followed by Staphylococcus saprophyticus. Trimethoprim-sulfamethoxazole has been the standard therapy for urinary tract infection; however, E. coli is becoming increasingly resistant to medications. Many experts support using ciprofloxacin as an alternative and, in some cases, as the preferred first-line agent. However, others caution that widespread use of ciprofloxacin will promote increased resistance.

[Epidemiology and etiology of urinary tract infections in the community. Antimicrobial susceptibility of the main pathogens and clinical significance of resistance].

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Urinary tract infections (UTI) are a frequent problem in primary care. They occur mainly in women without underlying diseases and with no functional or structural anomalies of the urinary tract; consequently most cases are considered uncomplicated UTI. Etiology is influenced by factors such as age, diabetes, spinal cord injury, urinary catheterization, and other factors. Escherichia coli causes 80-85% of acute episodes of uncomplicated cystitis. Staphylococcus saprophyticus, Proteus mirabilis, Streptococcus agalactiae and Klebsiella spp. are responsible for most of the remaining episodes. The spectrum of bacteria that causes complicated UTI is much broader. Rates of resistance have undergone considerable variations, and consequently the empirical treatment of UTI requires constant updating of the antibiotic sensitivity of the main uropathogens of the area, country or institution. To correctly interpret the global data on sensitivity, the type of UTI (uncomplicated versus complicated), sex, age and previous antibiotic therapy in each patient must be taken into account. Resistance in uncomplicated UTI has clinical significance (although less than in systemic infections such as bacteremia), which depends on whether the infection is cystitis or pyelonephritis.

Characteristics of Escherichia coli Urine Isolates and Risk Factors for Secondary Bloodstream Infections in Patients with Urinary Tract Infections.

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Escherichia coli is responsible for more than 80% of all incidences of urinary tract infections (UTIs). We assessed a total of 636 cases of patients with E. coli UTIs occurring in June 2019 in eight tertiary hospitals in South Korea for the traits of patients with E. coli UTIs, UTI-causative E. coli isolates, and risk factors associated with bloodstream infections (BSIs) secondary to UTIs. Antimicrobial susceptibility testing was conducted using the disc diffusion method, and the genes for extended-spectrum beta-lactamases (ESBLs) and plasmid-mediated ampC genes were screened by using PCR and sequencing. Multilocus sequence typing and virulence pheno-/genotyping were carried out. A total of 49 cases developed BSIs. The E. coli urine isolates primarily comprised sequence type 131 (ST131) (30.0%), followed by ST1193, ST95, ST73, and ST69. Three-quarters of the ST131 H30Rx isolates possessed the blaCTX-M-15-like gene, whereas 66% of H30R and 50% of H41 isolates possessed the blaCTX-M-14-like gene. All the ST1193 isolates showed biofilm formation ability, and three-quarters of the ST73 isolates exhibited hemolytic activity with high proportions of papC, focG, and cnf1 positivity. The prevalence of the ST131 H41 sublineage and its abundant CTX-M possession among the E. coli urine isolates were noteworthy; however, no specific STs were associated with bloodstream invasion. For BSIs secondary to UTIs, the papC gene was likely identified as a UTI-causative E. coli-related risk factor and urogenital cancer (odds ratio [OR], 12.328), indwelling catheter (OR, 3.218), and costovertebral angle tenderness (OR, 2.779) were patient-related risk factors. IMPORTANCE Approximately half of the BSIs caused by E. coli are secondary to E. coli UTIs. Since the uropathogenic E. coli causing most of the UTIs is genetically diverse, understanding the risk factors in the E. coli urine isolates causing the BSI is important for pathophysiology. Although the UTIs are some of the most common bacterial infectious diseases, and the BSIs secondary to the UTIs are commonly caused by E. coli, the assessments to find the risk factors are mostly focused on the condition of patients, not on the bacterial pathogens. Molecular epidemiology of the UTI-causative E. coli pathogens, together with the characterization of the E. coli urine isolates associated with the BSI secondary to UTI, was carried out, suggesting treatment options for the prevalent antimicrobial-resistant organisms.

Urinary tract infections.

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Urinary tract infections (UTIs) are still the precipitating cause for 7 million patient visits per year with total costs exceeding one billion dollars. Diagnostic modalities have become more "friendly" for the smaller laboratory with "dip stick" culture tests providing a rapid method of isolation of pathogens. In many cases, empiric therapy is more cost effective than culture in uncomplicated UTIs in women. The etiologic organisms implicated in UTIs have not changed dramatically over the past two decades, with E. coli still accounting for the majority of cases. Antibiotic susceptibility patterns have changed dramatically, with ampicillin losing utility die to the emergence of resistance. Quinolones, which have been exceedingly active against gram-negative enteric pathogens, are no longer universally active and more pathogenic organisms, such as pseudomonas, may be resistant. The emergence of other highly resistant organisms, such as Enterococcus faecium, must be watched for.

Maintained susceptibility to fosfomycin in extra-hospitalary urinary isolates of Escherichia coli. Strong association of fosfomycin resistance with age and ESBL production.

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Escherichia coli is isolated in most of uncomplicated community-acquired urinary tract infections (UTI) and fosfomycin is one of the treatments of choice. We analyzed the evolution of fosfomycin resistance in extrahospitalary E. coli urinary isolates and whether age and extended spectrum beta-lactamase (ESBL) production were associated to antibiotic resistance. A retrospective descriptive study was conducted from January 2017 to December 2022 including E. coli isolates from extrahospitalary urine samples. The susceptibility to fosfomycin remained above 95% during the study period. ESBL production and age above 80 years were significantly associated with increased fosfomycin resistance. We also analyzed the consumption of fosfomycin and it remained stable, although it was higher in the population >65 years. Greater resistance is observed in ESBL-producing strains and in patients over 65 years of age. A stable consumption of fosfomycin is associated with low resistance percentages maintained over the time.

Gender and age-dependent etiology of community-acquired urinary tract infections.

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Urinary tract infections (UTIs) are among the most frequent community-acquired infections worldwide. Escherichia coli is the most common UTI pathogen although underlying host factors such as patients' age and gender may influence prevalence of causative agents. In this study, 61 273 consecutive urine samples received over a 22-month period from outpatients clinics of an urban area of north Italy underwent microbiological culture with subsequent bacterial identification and antimicrobial susceptibility testing of positive samples. A total of 13 820 uropathogens were isolated and their prevalence analyzed according to patient's gender and age group. Overall Escherichia coli accounted for 67.6% of all isolates, followed by Klebsiella pneumoniae (8.8%), Enterococcus faecalis (6.3%), Proteus mirabilis (5.2%), and Pseudomonas aeruginosa (2.5%). Data stratification according to both age and gender showed E. coli isolation rates to be lower in both males aged ≥60 years (52.2%), E. faecalis and P. aeruginosa being more prevalent in this group (11.6% and 7.8%, resp.), as well as in those aged ≤14 years (51.3%) in whom P. mirabilis prevalence was found to be as high as 21.2%. Streptococcus agalactiae overall prevalence was found to be 2.3% although it was shown to occur most frequently in women aged between 15 and 59 years (4.1%). Susceptibility of E. coli to oral antimicrobial agents was demonstrated to be as follows: fosfomycin (72.9%), trimethoprim/sulfamethoxazole (72.9%), ciprofloxacin (76.8%), ampicillin (48.0%), and amoxicillin/clavulanate (77.5%). In conclusion, both patients' age and gender are significant factors in determining UTIs etiology; they can increase accuracy in defining the causative uropathogen as well as providing useful guidance to empiric treatment.

Detection of intracellular bacterial communities in human urinary tract infection.

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Urinary tract infections (UTIs) are one of the most common bacterial infections and are predominantly caused by uropathogenic Escherichia coli (UPEC). While UTIs are typically considered extracellular infections, it has been recently demonstrated that UPEC bind to, invade, and replicate within the murine bladder urothelium to form intracellular bacterial communities (IBCs). These IBCs dissociate and bacteria flux out of bladder facet cells, some with filamentous morphology, and ultimately establish quiescent intracellular reservoirs that can seed recurrent infection. This IBC pathogenic cycle has not yet been investigated in humans. In this study we sought to determine whether evidence of an IBC pathway could be found in urine specimens from women with acute UTI. We collected midstream, clean-catch urine specimens from 80 young healthy women with acute uncomplicated cystitis and 20 asymptomatic women with a history of UTI. Investigators were blinded to culture results and clinical history. Samples were analyzed by light microscopy, immunofluorescence, and electron microscopy for evidence of exfoliated IBCs and filamentous bacteria. Evidence of IBCs was found in 14 of 80 (18%) urines from women with UTI. Filamentous bacteria were found in 33 of 80 (41%) urines from women with UTI. None of the 20 urines from the asymptomatic comparative group showed evidence of IBCs or filaments.

Filamentous bacteria were present in all 14 of the urines with IBCs compared to 19 (29%) of 66 samples with no evidence of IBCs ($p < 0.001$). Of 65 urines from patients with E. coli infections, 14 (22%) had evidence of IBCs and 29 (45%) had filamentous bacteria, while none of the gram-positive infections had IBCs or filamentous bacteria. The presence of exfoliated IBCs and filamentous bacteria in the urines of women with acute cystitis suggests that the IBC pathogenic pathway characterized in the murine model may occur in humans. The findings support the occurrence of an intracellular bacterial niche in some women with cystitis that may have important implications for UTI recurrence and treatment.

Switch to amoxicillin-clavulanate oral therapy in urinary tract infection caused by extended-spectrum beta-lactamase-producing Escherichia coli: Assessment by chronic phase technetium-99m dimercaptosuccinic acid renal scintigraphy images.

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The incidence of urinary tract infection (UTI) caused by extensive beta-lactamase-producing Escherichia coli (ESBL-EC) is increasing, including in children. However, the available oral antibiotic treatment options for ESBL-EC are limited. Herein, we report the cases of two children diagnosed with UTI caused by ESBL-EC (ESBL-UTI) who were switched from empirical intravenous antibiotics in UTI to amoxicillin-clavulanic acid (AMPC/CVA) (14:1) after the causative organism was found to be ESBL-EC. A 3-month-old infant and an 8-month-old infant were admitted to our hospital with the chief complaint of fever. In both cases, UTI was suspected based on urinalysis results, and intravenous cefotaxime was started as an empiric antibiotic. In both cases, ESBL-EC was detected in urine culture, and the diagnosis of ESBL-UTI was confirmed. Results of antimicrobial susceptibility testing showed resistance to cefotaxime, but fever resolution was obtained in both cases following administration of intravenous cefotaxime. Since fever resolution was achieved, the antimicrobial was switched to oral AMPC/CVA (14:1) monotherapy with reference to antimicrobial susceptibility testing, and the two patients were discharged on days 5-6 of hospitalization. Antimicrobials were administered intravenously and orally for a total of 2 weeks. Chronic-phase technetium-99m dimercaptosuccinic acid renal scintigraphy showed no renal scarring. ESBL-UTI may require 2 weeks of intravenous antibacterial therapy, but in this case, both patients could be treated without renal scarring after conversion to oral AMPC/CVA alone. Since this is important to shorten the length of hospital stay, we will study the effect of this treatment modality in more cases in the future.

Urinary tract infections in women.

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Urinary tract infections (UTIs) are the most common infections seen in the hospital setting, and the second most common infections seen in the general population. Due to women's anatomy, UTIs are especially problematic for them, and up to one-third of all women will experience a UTI at some point during their lifetimes. Appropriate treatment of a UTI requires accurate classification that includes infection site, complexity of the infection, and the likelihood of recurrence. The predominant pathogen in both complicated and uncomplicated UTI remains pathogenic Escherichia coli, although Klebsiella sp. and Proteus appear with increased frequency in complicated UTI. Most often, bacteria cause UTIs by ascending means through the urethra into the bladder. Bacteria must possess virulence factors to cause

UTI. Host defense factors that predispose patients to UTI include urinary stasis, abnormal urinary tract anatomy, diabetes mellitus, debility, and aging. Estrogen-related issues and short urethras predispose women to UTI. Although urine culture, with >105 colony-forming units/mL (CFU/mL) in symptomatic patients, remains the diagnostic "gold standard," correlation of the patient's history and physical examination with urinalysis (including nitrite dipstick and leukocyte esterase test) results usually suffices to diagnose UTI. Three-day of antimicrobial treatment is recommended for simple cystitis. Acute pyelonephritis, an infection of the kidney parenchyma tissue, is treated with antibiotics for 7 to 14 days depending on the antimicrobial agent used and the severity of infection. In addition, patient classification determines the need for hospitalization or for urological imaging studies. Women with recurrent UTIs merit consideration for antimicrobial prophylaxis. Self-administered topical vaginal estradiol cream is an important adjunct in UTI prevention for postmenopausal women. Asymptomatic bacteruria only merits antimicrobial therapy in high-risk patients or those colonized with Proteus species.

The relative importance of Staphylococcus saprophyticus as a urinary tract pathogen: distribution of bacteria among urinary samples analysed during 1 year at a major Swedish laboratory.

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To determine the distribution of urinary tract pathogens with focus on Staphylococcus saprophyticus and analyse the seasonality, antibiotic susceptibility, and gender and age distributions in a large Swedish cohort. S. saprophyticus is considered an important causative agent of urinary tract infection (UTI) in young women, and some earlier studies have reported up to approximately 40% of UTIs in this patient group being caused by S. saprophyticus. We hypothesized that this may be true only in very specific outpatient settings. During the year 2010, 113,720 urine samples were sent for culture to the Karolinska University Hospital, from both clinics in the hospital and from primary care units. Patient age, gender and month of sampling were analysed for S. saprophyticus, Escherichia coli, Klebsiella pneumoniae and Proteus mirabilis. Species data were obtained for 42,633 (37%) of the urine samples. The most common pathogens were E. coli (57.0%), Enterococcus faecalis (6.5%), K. pneumoniae (5.9%), group B streptococci (5.7%), P. mirabilis (3.0%) and S. saprophyticus (1.8%). The majority of subjects with S. saprophyticus were women 15-29 years of age (63.8%). In this age group, S. saprophyticus constituted 12.5% of all urinary tract pathogens. S. saprophyticus is a common urinary tract pathogen in young women, but its relative importance is low compared with E. coli even in this patient group. For women in other ages and for men, growth of S. saprophyticus is a quite uncommon finding.